

RESEARCH NOTE

Development of a Self-Report Instrument for Exploring Team Factor, Leadership and Performance Relationships

Tudor Rickards, Ming-Huei Chen* and Susan Moger

Manchester Business School, Booth Street West, Manchester M15 6PB, UK and

*National Chi Nan University 1, University Road, Puli, Nantou Hsien, Taiwan 545

corresponding author email: trickards@man.mbs.ac.uk (T. Rickards)

Claims have been recently made for a seven-factor model that differentiated performance levels in project teams. We have tested the model with results from a self-report inventory. Forty-four opportunities for data collection were taken and a total of 1103 useable inventories collected from participants across European, Asian and African locations. Teams reported on were predominantly work-related. All seven factors correlate as predicted with leadership and team-performance criteria. We report results supporting the original proposition that team effectiveness, including its creativity, can be interpreted as arising through leadership interventions of a transformational kind, which impact on a set of theoretically-derived team factors.

Introduction

The persisting challenge of team dynamics

Gersick (1988) traces the development of team dynamics from early interest in the psychosocial and emotional aspects of teamwork, through the 1940s (Lewin, 1947), which later come to fruition in the group relations movements in Europe and America (Kleiner, 1996). The dynamics of team functioning has persisted as a topic of enduring interest for management researchers and practitioners alike, since the pioneering work of researchers such as Bales (1951); Bennis and Shepard (1956); and Homans (1951). Later influential contributions came from Hackman and co-workers (e.g. Hackman and Morris, 1975; Hackman, 1987); and Ancona and Caldwell (1992).

Practitioner-oriented treatments have suggested how member role preferences may contribute to

a team's performance (Belbin, 1981; Kirton, 1994; Larson and LaFasto, 1989; Tjosvold, 1992). Recent impetus to research on teams has grown with interest in the possibility of self-structuring teams (Manz and Sims, 1987) and cross-functional teams (Denison, Hart and Kahn, 1996). A related body of work grew up aimed at understanding the nature and impact of leadership on groups. As with the more general studies of teamwork, leadership research has evolved through various stages. These are generally considered to have departed from leadership as a trait, through style and contingency models. These studies empirically did not depart greatly from the classical dimensions of task and relationship-oriented behaviours until the arrival of the so-called new leadership studies, marked by emphasis on transformational leadership (Bryman, 1996). An important contribution to this development comes from Bass and co-workers (Bass, 1990; Bass and Avolio, 1994). Transformational leadership

opened up the possibility of exceptional performance that was hard to extract from earlier models.

One recurrent theme has been the development processes that occur within a team (Davies and Kuypers, 1985). There is a general consensus that such stages exist, albeit with uncertainties regarding the temporal and sequential details. A review of stage models by Tuckman (1965) became widely accepted as a summary of empirical observations and theoretical modelling (Hare, 1976; Mullins, 1989). Tuckman proposed the stages of form, storm, norm, perform, to which a final stage (adjourn) was subsequently added (Tuckman and Jensen, 1977). Nevertheless, various commentators cast doubt on the evidence that the model adequately captured the contingencies of team development (e.g. Huczynski and Buchanan, 1997; Rickards and Moger, 2000).

A modification to Tuckman's model, from a quite different conceptualization, has been proposed (Rickards and Moger, 1999, 2000) but as yet without quantitative exploration of its factors. The proposed framework was derived from studies of creative problem-solving teams, and their leadership processes. These experiences were related to the Tuckman sequences by the suggestion that some teams were never observed to pass through the storm stage, and that other teams were found which had exceeded prior expectations and norms. In this model, performance was associated with an enabling ('creative') leadership style, which deliberately introduces structures supportive of norm-challenging team behaviours. A set of team factors was proposed derived from the creative leadership efforts. The published reports of the framework have been largely speculative, and supported only by simple qualitative and numerical observations. The researchers have recognized the benefits that would derive from the development of a self-report instrument to support quantitative examination of the framework.

The team factors influenced by creative leadership

The provenance, and a further explanation of the team factors, can be found in Rickards and Moger (2000). A synoptic account of the factors, leadership styles and performance variables is shown in Table 1.

Objectives of the research

The longer-term objective of the research is to provide means of supporting creative activities at the level of project teams, grounded in a currently under-explored theory of team development. A step towards this would be the development of a diagnostic instrument based on the proposed theory. The instrument is intended for use across the widest possible range of project teams, and applied rapidly and with confidence in its reliability and validity. The researchers examined a range of instruments, and concluded that no prevailing instrument would be at the same time simple to use, and of adequate reliability and validity.

Methods

Samples

Data were collected by convenience sampling at all opportunities within a wide range of training and action research programmes over the period May 1998–September 1999. The occasions were all broadly dealing with organizational creativity and innovation. Respondents supplied information from teams within their experience, and went far beyond activities connected with specific creative problem-solving methodologies. Forty-four opportunities for data collection occurred. A total of 1103 useable inventories were collected from UK, European (outside of UK), Asian and African locations. Data were stored, checked for input errors (inventories with more than five missing values were eliminated) and analysed using SPSS. Three hundred and eighty three participants (34.7%) were British; 249 participants (22.6%) were Asian and 204 (18.5%) were Northern African. Only 44 participants (4%) were European (non-British). Two hundred and twenty three (20.2%) were participants in MBA programmes coming from a variety of countries. Details of country data, respondent age ranges and gender are available from the authors.

Development of the team factor inventory (TFI)

In its current format, the inventory collects self-reports on seven sets of three items for the seven team factors. This represents the core inventory. Additional items are seen as optional elements incorporated to meet requirements for specific

Table 1. Key features of team factors scales, leadership and output variables

Team factor inventory (TFI) scales	Key features
<i>Seven team factors</i>	
Platform of understanding	Team members understand and respect each other's viewpoints and the team shares knowledge, beliefs, and assumptions. These elements comprise a 'platform of understanding' from which new ideas develop.
Shared vision	Team members share a sense of purpose and responsibility that motivates and sustains team progress.
Creative climate	Team members trust each other, and share a positive and supportive approach to stimulate creativity in work situations.
Idea owners	The ideas that are given more attention are those perceived as open to strong sponsorship by team members. A creative team generates and sustains new and valued ideas on task-related issues, in a manner supportive of the behavioural needs of its members.
Resilience	Team members are flexible as they hit setbacks and frustration.
Network activators	Team members are good at networking with key individuals outside formal organizational systems, exchanging ideas and offering mutual support.
Learning from experience	Team members are oriented toward learning from their experiences, thus permitting growth, change, adaptation, and creative problem-solving.
<i>Team leadership</i>	
Transformational leadership	The team leader tends to be inspirational, relying on personal charisma and on gaining trust in leader's abilities, and empowering team members, who act out of self-generated enthusiasm and interests.
Transactional leadership	The team leader tends to keep the team focus on tasks necessary for success, using implicit and explicit instrumental rewards as motivational and control mechanisms.
<i>Performance criteria</i>	
Creativity	A creative team creates new ideas on task-related issues and displays imagination at work.
Productivity	A productive team achieves its intended goals (task and team goals).

investigations. In the work reported here, we also collected data on two items for transformational leadership and one item each for transactional leadership, creative output and productivity.

A defining statement had been developed for each of the team factors in the earlier stage of the research. Three to four simpler items were constructed from each of the defining statements. The set of items was discussed with MBAs from various cultural backgrounds, a process that led to selection of three items, (in a few cases with suggested clarifications) representing each of the seven factors. The items were believed by the researchers to capture the essential aspects contained in the definitions.

A five-point scaling method was chosen to measure each item (strongly agree . . . strongly disagree). Unintentionally, this and other design decisions were similar to those reached by Sutton and Hargadon (1996), who also combined

qualitative and simple quantitative measures in their investigation of the effectiveness of a highly creative design organization. Negative sense items have been used by other researchers to explore barriers or impediments within team-level measures (e.g. Amabile *et al.*, 1996). The inclusion of negative sense factors was postponed for future studies requiring investigation of barriers to team performance.

Results

Normative samples

Normative data on the TFI were collected from three coherent categories of organizations, representing a variety of functional departments within those organizations. Two hundred and fifty-four participants (23%) were from large public groups;

188 participants (17.1%) were from large private groups, and 127 participants (11.5%) were from SMEs. Samples including more than one of these categories were combined into a miscellaneous group of 534 participants (48.4%). Table 2 summarizes means and standard deviations for the available data.

Reliability of TFI scales

In Table 3, the final versions of the team-factor items are shown, together with self-scale/other-scale correlations. The results show that all items correlate more strongly (higher than 0.70) with items within their own scales than with items from other scales. For completeness, the final versions of the leadership and output items have also been included in Table 3.

A factor analysis (principal components) with varimax rotation was carried out with 20 of the 21 items, to assess the underlying structure of the TFI items. Item TF13 from the idea-owners scale had been eliminated from this analysis since it did not fit our criterion for preserving it and it did not have its primary loading on the same factor as the other two items from the same scale.

The most interpretable pattern of factors was obtained when the number of factors was limited to six (Table 4). One factor contained the complete sets of three items for platform of understanding and for climate. The other five factors corresponded to the five residual team factors (three items for each factor except the idea-owners

factor, now containing two items, as explained above). The eigenvalue on each factor is higher than one, and the percentage of cumulative variance explained achieves 70%.

Although the items of 'platform of understanding' scale and the items of 'climate' scale loaded on a single factor, these two scales represent potentially distinct modes whereby leaders are believed to influence team dynamics. Therefore, the six items (TF01, TF02, TF03, TF07, TF08 and TF09) were refactored and rotated in order for them to be optimally interpretable (Carmines and Zeller, 1979).

Table 5 examines zero-order correlations between team factors, leadership scales and performance scales. Of the 55 intercorrelations, each scale correlated moderately with all other scales ($p < 0.01$). Team factors correlated more highly with transformational leadership (0.42–0.59) than with transactional leadership (0.33–0.45); and correlated more highly with creativity (0.48–0.73) than with productivity (0.42–0.63). Transformational leadership correlated 0.48 with transactional leadership and creativity correlated 0.71 with productivity. Transformational leadership correlated more highly with creativity (0.65) and productivity (0.55) than did transactional leadership (0.46 for creativity and 0.47 for productivity).

The scale reliabilities across seven team factors range from 0.73 to 0.81. The scale reliability is 0.89 for the transformational leadership scale.

Table 2. Means and standard deviations in the TFI for UK samples

	SMEs	Large private	Large public	Miscellaneous	Total
Team factor scales					
Platform of understanding	3.45 (0.77)	3.43 (0.80)	3.22 (1.01)	2.92 (0.99)	3.31 (0.87)
Shared vision	3.60 (0.82)	3.56 (0.95)	3.33 (0.98)	3.22 (1.16)	3.48 (0.97)
Climate	3.67 (0.74)	3.66 (0.94)	3.41 (0.85)	3.06 (1.05)	3.51 (0.93)
Resilience	3.83 (0.75)	3.78 (0.78)	3.76 (0.70)	3.19 (0.98)	3.68 (0.84)
Idea owners	3.46 (0.91)	3.44 (0.98)	3.24 (0.88)	2.95 (1.02)	3.33 (0.97)
Network activators	3.48 (0.88)	3.47 (0.89)	3.45 (0.80)	3.04 (0.99)	3.38 (0.91)
Learning from experience	3.58 (0.82)	3.48 (0.85)	3.32 (1.03)	3.00 (1.02)	3.40 (0.92)
Leadership variables					
Transformational leadership	3.38 (0.88)	3.13 (1.10)	2.91 (1.19)	2.69 (1.11)	3.10 (1.07)
Transactional leadership	3.66 (0.97)	3.65 (0.92)	3.76 (1.17)	3.51 (1.22)	3.64 (1.03)
Performance criteria					
Creativity	3.70 (0.86)	3.46 (0.94)	3.41 (0.93)	3.11 (1.13)	3.46 (0.98)
Productivity	3.87 (1.00)	3.80 (0.94)	3.53 (1.18)	3.23 (1.33)	3.68 (1.10)
Sample size	127	140	38	78	383

Note: The numbers in parentheses show standard deviations.

Table 3. Item descriptions, and correlation analysis (self-scale/other-scale) on team factor items

Items	Correlations	
	Self scale	Other scales
Platform of understanding		
TF01 The team worked successfully at understanding each other's job requirements	0.76	0.57
TF02 The team worked successfully at understanding each other's personal needs	0.76	0.57
TF03 Team members developed a common understanding of situational requirements	0.71	0.60
Shared vision		
TF04 The team tended to have a shared view ('vision') of team purpose	0.84	0.59
TF05 The team had strong loyalty to its team purpose	0.84	0.66
TF06 Team members had no confusions over its team purpose	0.82	0.52
Creative climate		
TF07 The team climate was warm and positive	0.80	0.60
TF08 The team environment was interesting and challenging	0.71	0.57
TF09 The team members trusted and supported one another	0.81	0.62
Resilience		
TF10 If we hit an unexpected problem we pulled together to deal with it	0.81	0.63
TF11 If we hit a problem we had confidence in our own abilities to sort things out	0.79	0.57
TF12 We were able to bounce back after any setback to our plans	0.76	0.56
Idea owners		
TF13 People often suggested ideas and then ran with them	0.72	0.37
TF14 There were willing volunteers to try out new ideas	0.80	0.52
TF15 People were prepared to take responsibilities for making new ideas work	0.79	0.60
Network activators		
TF16 Team members had contacts outside the team that were helpful	0.83	0.35
TF17 Team members were good at mobilizing help from outside the team	0.88	0.43
TF18 Team members were good networkers	0.71	0.52
Learning from experience		
TF19 We were good at learning from mistakes	0.79	0.62
TF20 We talked things over when things went wrong	0.82	0.54
TF21 We tended to try out new ideas after things went wrong	0.79	0.51

Notes: (n = 1103). All correlations are statistically significant at the $p < 0.001$ level. The inventory also included two items examining transformational leadership (Team leadership tended to be inspirational; Team leadership tended to be motivational), and an item respectively on transactional leadership (Team leadership tends to be results oriented); creativity (We were a creative team); and productivity (We were a productive team).

Validity of TFI scales

A stepwise multiple regression analysis was applied to predict creativity and productivity from the team-factors scales. Collinearity was diagnosed by examining the tolerance statistic (range 0–1) where 0 represents an undesirable linear combination of variables. The value of tolerance ranged from 0.37 to 0.69. This indicated that the interpretation of the regression coefficients was not affected seriously by multi-collinearity (Hair *et al.*, 1998). The final fit models of predictors of creativity and productivity are shown in Table 6.

Six factors (all except the 'resilience' factor) were significantly loaded as predictors of creativity. In the model of productivity prediction, all seven

factors were significantly loaded to predict productivity. The team factors scales were significantly associated with team outputs (adjusted $R^2 = 0.67$ for creativity predictions and $R^2 = 54$ for productivity predictions).

Conclusions

The current study reports the construction of a simple-to-use, team-factor inventory for investigations of the dynamics of team performance, with particular emphasis on non-routine ('creative') outputs. The extended process of item analysis (self scale/other scales), factor analysis, scale reliability (internal consistency), test-retest reliabilities, and predictive validities has begun, as

Table 4. Results of factor analysis of team factors

Items	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
Platform of understanding						
TFI 02	0.70	0.18	0.19	0.23	0.10	0.24
TFI 01	0.67	0.11	0.43	0.14	0.05	0.15
TFI 03	0.43	0.14	0.42	0.30	0.23	0.18
Creative climate						
TFI 09	0.73	0.16	0.21	0.15	0.32	0.14
TFI 07	0.72	0.15	0.19	0.19	0.25	0.12
TFI 08	0.55	0.16	0.12	0.14	0.36	0.21
Network activators						
TFI 17	0.15	0.87	0.13	0.13	0.08	0.14
TFI 16	0.11	0.86	0.07	0.12	0.07	0.05
TFI 18	0.21	0.62	0.25	0.10	0.19	0.25
Shared vision						
TFI 06	0.24	0.15	0.80	0.10	0.18	0.07
TFI 04	0.28	0.18	0.69	0.19	0.16	0.23
TFI 05	0.46	0.16	0.52	0.20	0.31	0.10
Learning from experience						
TFI 20	0.27	0.10	0.13	0.81	0.18	0.06
TFI 21	0.13	0.12	0.10	0.79	0.17	0.23
TFI 19	0.31	0.22	0.30	0.57	0.22	0.10
Resilience						
TFI 11	0.23	0.10	0.28	0.14	0.70	0.24
TFI 12	0.18	0.19	0.13	0.37	0.66	0.09
TFI 10	0.51	0.07	0.18	0.25	0.52	0.12
Idea owners						
TFI 14	0.23	0.14	0.14	0.17	0.13	0.83
TFI 15	0.29	0.25	0.18	0.18	0.25	0.70
Eigenvalue	3.56	2.31	2.25	2.25	1.92	1.66
Cumulative percentage variance	17.81	29.37	40.64	51.88	61.47	69.79

Notes: n = 1103.

The numbers TF01, TF02 etc represent the coding numbers of items shown in Table 3.

Extraction method: Principal component analysis. Rotation method: varimax.

reported above. The instrument provides preliminary support for a modification by Rickards and Moger (2000) to Tuckman's model of team development (Tuckman, 1965; Tuckman and Jensen, 1977). Transformational leadership (Bass and Avolio, 1994) has been shown to be associated with seven predicted team factors, and with output variables designated as creativity and productivity.

We acknowledge that in this study, we are taking self-report evidence of inventory items as our sole, surrogate measure of these constructs. Taking these considerable qualifications into account, we consider that the results are promising. Correlation analysis suggests that all seven team factors significantly correlate with each other and correlate positively with creativity and productivity ($p < 0.01$). Thus, the accumulating evidence encourages us to believe that the results go beyond those expected from a single-factor

explanation, for example through a gross centralizing tendency of self-reports. We have set out on a path to add to knowledge of factors influencing team effectiveness, particularly of outstanding and creative effectiveness. The instrument, in current and future extended forms, offers promise at the start of such a journey.

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Table 5. Correlations and reliabilities for the TFI scales

The TFI scales	1	2	3	4	5	6	7	8	9	10	11
1. Platform of understanding	(0.76)										
2. Shared vision	0.68	(0.78)									
3. Climate	0.71	0.67	(0.79)								
4. Resilience	0.63	0.62	0.68	(0.73)							
5. Idea owners	0.53	0.51	0.57	0.55	(0.74)						
6. Network activators	0.45	0.45	0.43	0.40	0.46	(0.81)					
7. Learning from experience	0.58	0.56	0.57	0.63	0.49	0.42	(0.77)				
8. Transformational leadership	0.57	0.57	0.59	0.51	0.49	0.42	0.54	(0.89)			
9. Transactional leadership	0.36	0.41	0.39	0.41	0.36	0.33	0.45	0.48	+		
10. Creativity	0.64	0.67	0.73	0.62	0.61	0.48	0.60	0.65	0.46	+	
11. Productivity	0.62	0.61	0.63	0.61	0.51	0.42	0.55	0.55	0.47	0.71	+

Note: n = 1103.

Scale reliabilities are shown along the diagonal.

+ There is only one item in the scale. No reliability calculation is possible.

All correlations are statistically significant at the $p < 0.01$ level.

Table 6. Results of multiple regression analysis: team factors as predictors of creativity and productivity

Team-factor scales	Predictors of creativity			Predictors of productivity		
	B	β	t	B	β	t
Constant	0.17			0.05		
Platform of understanding	0.10	0.09	2.71**	0.20	0.16	4.28***
Shared vision	0.22	0.21	6.85***	0.19	0.17	4.65***
Creative climate	0.34	0.31	9.73***	0.19	0.17	4.24***
Resilience	+			0.22	0.17	4.69***
Idea owners	0.19	0.19	7.15***	0.07	0.06	1.99*
Network activators	0.07	0.07	2.85**	0.06	0.06	2.04*
Learning from experience	0.15	0.14	5.18***	0.12	0.10	3.20***
Adjusted R ²	0.67			0.54		

Notes: Stepwise method is applied for variable selection.

B represents unstandardized beta coefficients; β represents standardized beta coefficients.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

+ non-significant predictor.

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